HOME ASSIGNMENT 4
ID : 12341550
DATE : 18/10/25

Q1:

code :

```
#include<stdio.h>
#include<pthread.h>
#include<stdlib.h>
#define SEGMENT_SIZE 10
#define SIZE 100

typedef struct
{
    int *arr;
    int start;
}args;

void *partial_sum(void *arg)
{
    args *a = (args *)arg;
    int sum = 0;
    for (int i = a->start; i<a -> start + SEGMENT_SIZE; ++i)
        sum += a->arr[i];
    int *result = malloc(sizeof(int));
    *result = sum;
    return result;
}

void *total_sum(void *arg){
    int * partial_sums = (int *)arg;
    int total = 0;
    for(int i = 0; i < SEGMENT_SIZE; i++){
```

```
        total += partial_sums[i];
    }
    int * result = malloc(sizeof(int));
    *result = total;
    return result;
}

int main(){
    int arr[SIZE];
    for(int i = 0; i < SIZE; i++) arr[i] = i + 1;

    pthread_t thread[SEGMENT_SIZE];
    args segment[SEGMENT_SIZE];
    int segment_sum[SEGMENT_SIZE];

    for(int i = 0; i < 10; i++){
        segment[i].arr = arr;
        segment[i].start = i * SEGMENT_SIZE;
        pthread_create(&thread[i],NULL,partial_sum,&segment[i]);
    }

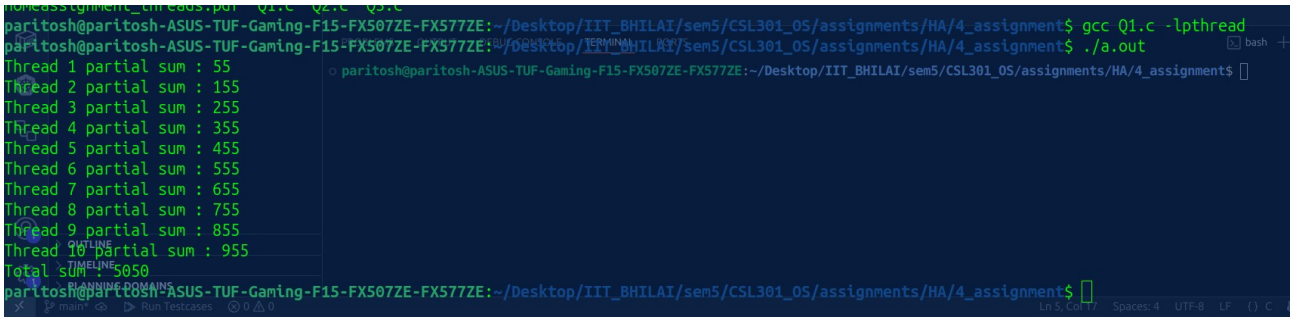

    for(int i = 0; i < SEGMENT_SIZE; i++){
        void * result;
        pthread_join(thread[i],&result);
        segment_sum[i] = *((int *)result);
        printf("Thread %d partial sum : %d\n",i + 1,*((int *)result));
        free(result);
    }

    pthread_t master_sum;
    void *total;

    pthread_create(&master_sum,NULL,total_sum,segment_sum);
    pthread_join(master_sum,&total);
```

```
    printf("Total sum : %d\n",*((int *)total));
}
```

OUTPUT:



Q2:

code:

```
#include<stdio.h>
#include<pthread.h>
#include<stdlib.h>
#define SEGMENT_SIZE 10
#define SIZE 100

typedef struct
{
    int *arr;
    int start;
}args;

void *partial_sum(void *arg)
{
    args *a = (args *)arg;
```

```
    int sum = 0;
    for (int i = a->start; i<a -> start + SEGMENT_SIZE; ++i)
        sum += a->arr[i];
    int *result = malloc(sizeof(int));
    *result = sum;
    return result;
}

void *total_sum(void *arg){
    int * partial_sums = (int *)arg;
    int total = 0;
    for(int i = 0; i < SEGMENT_SIZE; i++){
        total += partial_sums[i];
    }
    int * result = malloc(sizeof(int));
    *result = total;
    return result;
}

int main(){
    int arr[SIZE];
    for(int i = 0; i < SIZE; i++) arr[i] = i + 1;

    pthread_t thread[SEGMENT_SIZE];
    args segment[SEGMENT_SIZE];
    int segment_sum[SEGMENT_SIZE];

    for(int i = 0; i < 10; i++){
        segment[i].arr = arr;
        segment[i].start = i * SEGMENT_SIZE;
        pthread_create(&thread[i],NULL,partial_sum,&segment[i]);
    }


    for(int i = 0; i < SEGMENT_SIZE; i++){
```

```
        void * result;
        pthread_join(thread[i],&result);
        segment_sum[i] = *((int *)result);
        printf("Thread %d partial sum : %d\n",i + 1,*((int *)result));
        free(result);
    }

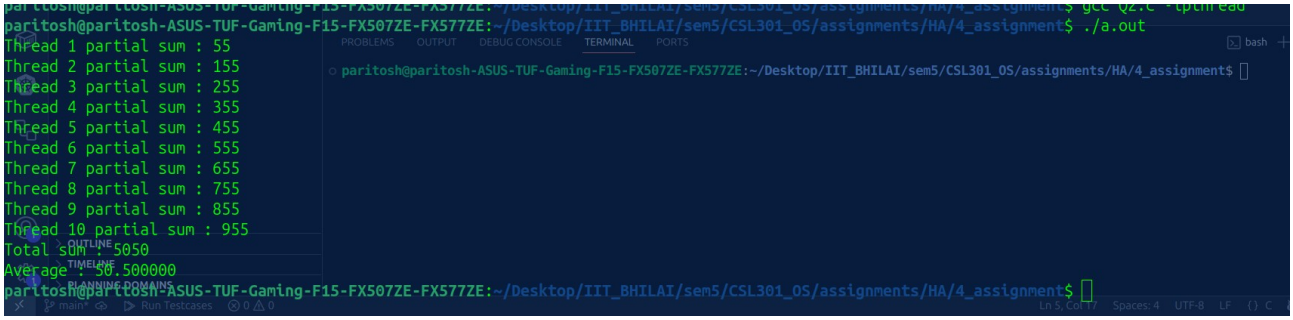
    pthread_t master_sum;
    void *total;

    pthread_create(&master_sum,NULL,total_sum,segment_sum);
    pthread_join(master_sum,&total);

    printf("Total sum : %d\n",*((int *)total));

    double average = (double)*((int *)total) / SIZE;
    printf("Average : %f\n",average);
}
```

OUPUT:



Q3:

code:

```
#include <stdio.h>
#include <pthread.h>
#include <stdlib.h>
#include <time.h>
#include <limits.h>

#define N 10
#define SEGMENT_SIZE 10
#define SIZE (N * SEGMENT_SIZE)

typedef struct
{
    int *arr;
    int start;
} args;

void *partial_max(void *arg)
{
    args *a = (args *)arg;
    int max_val = INT_MIN;

    for (int i = a->start; i < a->start + SEGMENT_SIZE; ++i)
    {
        if (a->arr[i] > max_val)
        {
            max_val = a->arr[i];
        }
    }

    int *result = malloc(sizeof(int));
    *result = max_val;
    return result;
}

void *overall_max(void *arg)
```

```
{
    int *partial_maxima = (int *)arg;
    int total_max = INT_MIN;

    for (int i = 0; i < N; i++)
    {
        if (partial_maxima[i] > total_max)
        {
            total_max = partial_maxima[i];
        }
    }

    int *result = malloc(sizeof(int));
    *result = total_max;
    return result;
}

int main()
{
    srand(time(NULL));

    int arr[SIZE];
    printf("Array: [");
    for (int i = 0; i < SIZE; i++)
    {
        arr[i] = rand() % 100;
        printf("%d%s", arr[i], (i == SIZE - 1) ? "" : ", ");
    }
    printf("]\n\n");

    pthread_t thread[N];
    args segment[N];
    int segment_max[N];

    for (int i = 0; i < N; i++)
```

```
    {
        segment[i].arr = arr;
        segment[i].start = i * SEGMENT_SIZE;
        pthread_create(&thread[i], NULL, partial_max,
&segment[i]);
    }

    for (int i = 0; i < N; i++)
    {
        void *result;
        pthread_join(thread[i], &result);

        segment_max[i] = *((int *)result);

        printf("Thread %d max: %d\n", i + 1, *((int *)result));

        free(result);
    }

    pthread_t master_max;
    void *overall;

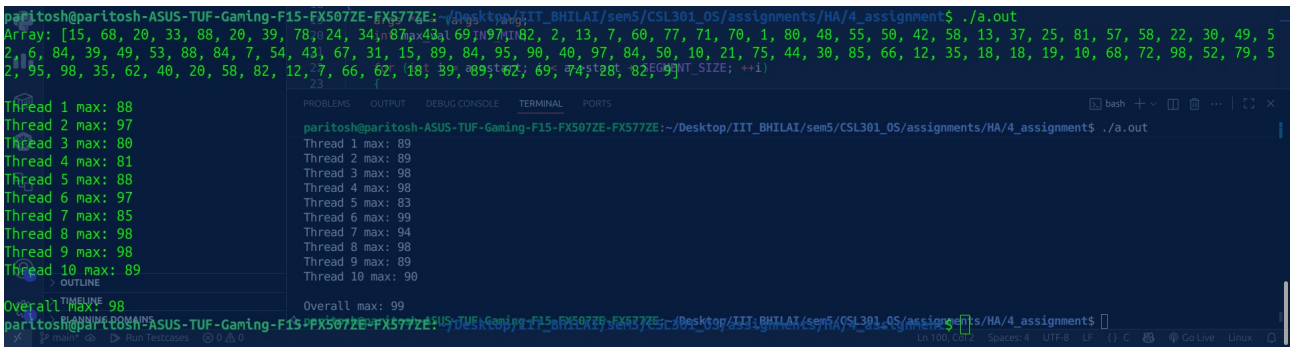
    pthread_create(&master_max, NULL, overall_max,
segment_max);
    pthread_join(master_max, &overall);

    printf("\nOverall max: %d\n", *((int *)overall));

    free(overall);

    return 0;
}
```

OUTPUT:


paritosh@paritosh-ASUS-TUF-Gaming-F15-FX507ZE-FX577ZE:~/Desktop/IIT_BHILAI/sem5/CSL301_OS/assignments/HA/4_assignment$ ./a.out
Array: [15, 68, 20, 33, 88, 20, 39, 78, 24, 34, 87, 43, 69, 97, 82, 2, 13, 7, 60, 77, 71, 70, 1, 80, 48, 55, 50, 42, 58, 13, 37, 25, 81, 57, 58, 22, 30, 49, 52, 6, 84, 39, 49, 53, 88, 84, 7, 54, 43, 67, 31, 15, 89, 84, 95, 90, 40, 97, 84, 50, 10, 21, 75, 44, 30, 85, 66, 12, 35, 18, 18, 19, 10, 68, 72, 98, 52, 79, 52, 95, 98, 35, 62, 40, 20, 58, 82, 12, 7, 66, 62, 18, 39, 89, 62, 69, 74, 28, 82, 9]
Thread 1 max: 88
Thread 2 max: 97
Thread 3 max: 80
Thread 4 max: 81
Thread 5 max: 88
Thread 6 max: 97
Thread 7 max: 85
Thread 8 max: 98
Thread 9 max: 98
Thread 10 max: 89
Overall max: 98
paritosh@paritosh-ASUS-TUF-Gaming-F15-FX507ZE-FX577ZE:~/Desktop/IIT_BHILAI/sem5/CSL301_OS/assignments/HA/4_assignment$
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
bash
paritosh@paritosh-ASUS-TUF-Gaming-F15-FX507ZE-FX577ZE:~/Desktop/IIT_BHILAI/sem5/CSL301_OS/assignments/HA/4_assignment$ ./a.out
Thread 1 max: 89
Thread 2 max: 89
Thread 3 max: 98
Thread 4 max: 98
Thread 5 max: 83
Thread 6 max: 99
Thread 7 max: 94
Thread 8 max: 98
Thread 9 max: 89
Thread 10 max: 90
Overall max: 99
paritosh@paritosh-ASUS-TUF-Gaming-F15-FX507ZE-FX577ZE:~/Desktop/IIT_BHILAI/sem5/CSL301_OS/assignments/HA/4_assignment$
OUTLINE
TIMELINE
PLANNING DOMAINS
main*
Run Testcases
Ln 100, Col 2
Spaces: 4
UTF-8
LF
C
Go Live
Linux